CTSpinoPelvic1K: a fused spine and pelvis CT segmentation dataset built for lumbosacral transitional anatomy

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(Dated: 23 August 2026)

Purpose: Lumbar levels cannot be numbered reliably when a lumbosacral transitional vertebra (LSTV) is present, because sacralisation and lumbarisation are one morphology under two counts and a spine-limited field of view cannot distinguish them. Public segmentation collections annotate the spine or the pelvis, rarely both, and none annotate the structures that make the count decidable. CTSpinoPelvic1K provides 802 CT records with a unified spine, pelvis, rib and femoral segmentation designed so that level assignment is anchored at both ends of the lumbar column.

Acquisition and Validation Methods: Records were built by patient-level crosswalk of three public collections — TCIA CT COLONOGRAPHY imaging, CT-Spine1K[3] vertebral labels, and CTPelvic1K[4] pelvic labels — with labels placed on the acquisition carrying the highest bone coverage and unified under a VerSe-native scheme. Validation is threefold: per-case geometric invariants (802/802 pass); rib–vertebra incidence across 5,749 evaluable ribs (2 residual offsets, 0.035%); and agreement of derived spinopelvic measures with published reference values.

Data Format and Usage Notes: Gzipped NIfTI image/label pairs sharing an identical affine, canonicalised to PIR, with frozen patient-grouped LSTV-stratified five-fold splits and a reference loader. Archived at 10.5281/zenodo.XXXXXXX.

Potential Applications: Segmentation and level-numbering benchmarks, anatomical variant studies, surgical planning, and opportunistic screening research. *Limitations:* thoracic ground truth is field-of-view limited; postural angles are supine; soft-tissue and hardware classes are declared but unpopulated; transitional labels derive from more than one source and are not uniformly adjudicated.

I. PURPOSE

Naming a lumbar vertebra requires counting, and counting requires two fixed ends. In the presence of a lumbosacral transitional vertebra neither end is where it is assumed to be: the same bone is a sacralised L5 or a lumbarised S1 depending only on where the count starts, and a field of view that shows the lumbar spine without the thoracolumbar junction cannot decide between them. LSTV is common — reported between 4% and 30% depending on definition — and wrong-level surgery is its most serious consequence.

Existing public collections do not make the problem tractable. Spine collections annotate vertebrae without the pelvis; pelvic collections annotate the sacrum and ilia without the vertebral count; neither annotates ribs, which are the rostral anchor that makes the count decidable. CTSpinoPelvic1K exists to put all of it in one coordinate frame.

The design commitment is that *the dataset never asserts a level name it cannot support*. Where the field of view cannot decide, the release reports count-free quantities: an interval count between the lowest rib and the sacrum, a rib length as a fraction of the rib above it, a transverse process span and its gap to the ala.

II. ACQUISITION AND VALIDATION METHODS

II.A. Sources and patient-level crosswalk

Each patient in TCIA CT COLONOGRAPHY was scanned prone and supine. Vertebral labels (CTSpine1K) and pelvic labels (CTPelvic1K) were placed independently onto whichever acquisition carried the greatest bone coverage. For most patients both land on the same series (*fused*); for the remainder they do not, and the patient contributes a *spine_only* and a *pelvic_native* record. Of 802 records, 440 are spine_only, 342 fused and 20 pelvic_native.

Vertebral labels are radiologist ground truth throughout; no pseudolabelled spine is shipped. The pelvis is real where fused and pseudolabelled elsewhere, with quality reported as held-out Dice on the pelvic_native scans.

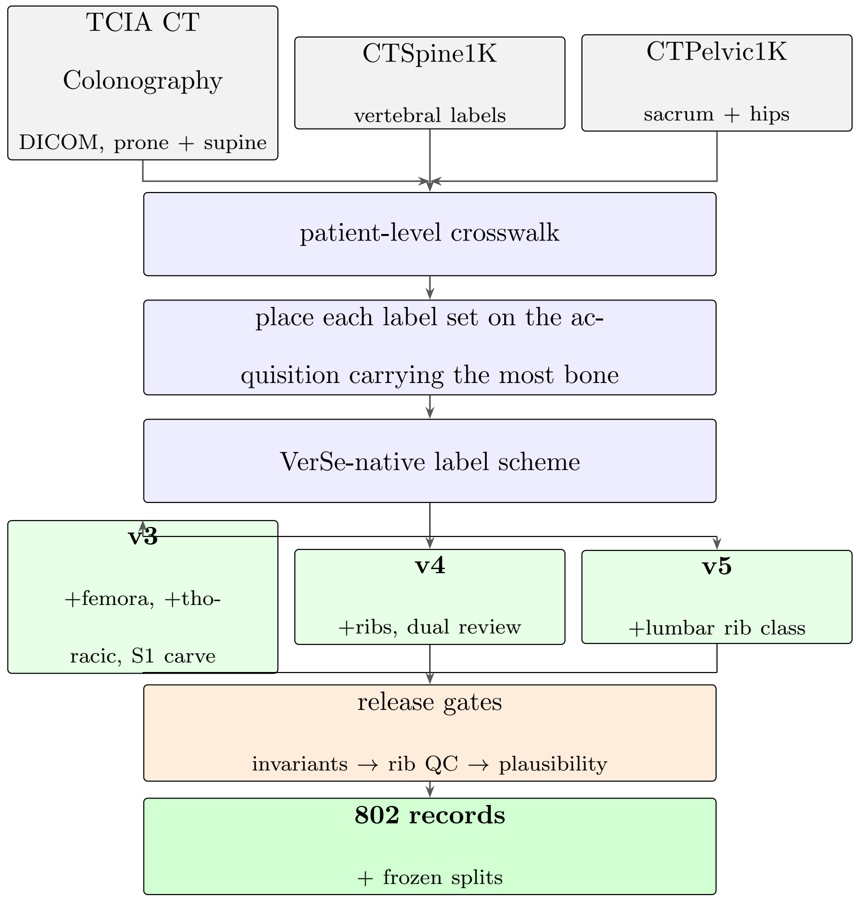
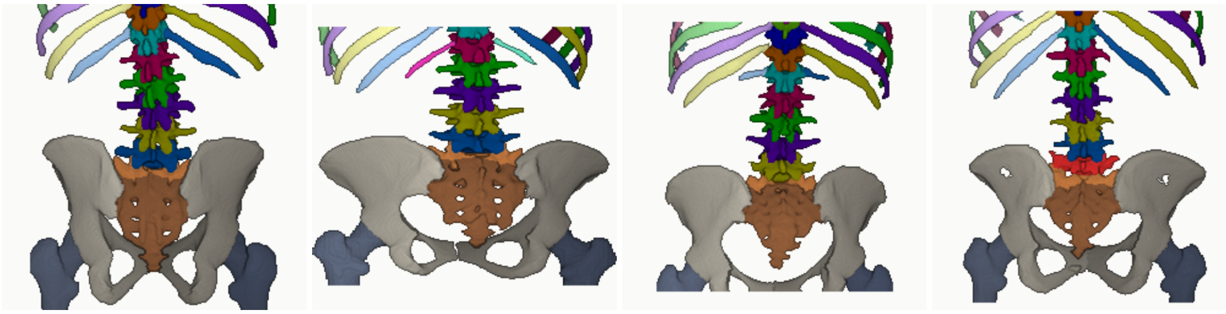


Fig. 1 Construction. Three public collections are joined at the patient level; each label set is placed on the acquisition carrying the most bone, so spine and pelvic labels sometimes land on different acquisitions of the same patient and are exported as separate records. Release gates run *before* any derived measurement: a corpus that has not established internal consistency produces numbers that look finished rather than broken.

II.B. Label scheme and counting anchors

The scheme is VerSe-native: vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12 = 8–19, L1–L6 = 20–25, sacrum 26), and every non-VerSe structure takes a fixed identifier above that range — S1 carved from the sacrum (29), hips (30–31), femora (32–33), ribs (34–45 left, 46–57 right), soft tissue (58–73), lumbar ribs (74–75) and hardware (76–79).

Two anchors make the count decidable. Rostrally, **T12** is the last rib-bearing vertebra and sits directly above ground-truth L1. Caudally, **S1** is carved as a distinct class, giving the bottom bracket and the sacral endplate landmark required for sacral slope and pelvic



(a) five rib-free (typical) (b) four, with a lumbar rib (c) four, no lumbar rib (d) six rib-free

FIG. 2 The configurations the corpus exists to represent, rendered from the released labels at one shared posterior viewing angle. Columns are captioned by the number of rib-free vertebrae between the lowest rib-bearing vertebra and the sacrum, not by a level name: a spine-limited field of view cannot settle whether (b) is a sacralised L5 or (d) a lumbarised S1, and the count can be checked against the picture where a name cannot. Panels (b) and (c) carry the same count, reached from opposite ends — in (b) a rib on the first lumbar-type vertebra shortens the interval from above, in (c) the lowest lumbar segment is incorporated into the sacrum and shortens it from below. Surfaces and colours are those of the released label descriptor, so the claims are checkable against the image: the lumbar rib in (b) carries its own class and renders magenta, and the sixth lumbar vertebra in (d) renders red directly above the sacrum. Each column is cropped to a fixed height above the top of the sacrum rather than to a fraction of the labelled spine, so the columns are framed on the same anatomy and a taller patient shows fewer levels rather than a rescaled skeleton.

incidence. With both brackets present, L5 and L6 are distinguishable where a spine-limited view alone would not be.

A rib on a lumbar body receives its own class rather than being forced to be a twelfth rib. A lumbar rib and a hypoplastic twelfth rib are the same object under two counts; assigning it a number would bake one reading into the label. 16 records carry one.

Figure 2 shows why the count is the primitive and the name is derived. Of the 29 records with four rib-free vertebrae, 15 reach that count through a lumbar rib and 14 without one; the two are indistinguishable by the count alone and are distinguished here only because rib presence is itself labelled.

II.C. Validation

Validation is layered, and the ordering matters: measurements computed on a corpus that has not established internal consistency do not look broken, they look finished.

Geometric invariants. Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for left- and right-sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed. This check exists because a renumbering pass once wrote a reoriented array under a canonical affine and transposed one label away from its image — a failure invisible to every downstream step.

Rib–vertebra incidence. Each rib is matched to its nearest vertebra and compared against the vertebra its number implies. Of 11,548 ribs present, 5,788 are not evaluable at all — the vertebra their number implies lies outside the field of view and nothing else is within the anchor distance — and a further 11 have no vertebra in reach. That leaves 5,749 with an anchor to check, of which two remain offset (0.035%). The denominator is stated because quoting two against 11,548 halves the rate by counting ribs the check never examined,

TABLE I Release validation. Gates run in order and a failure stops the chain.

Check	Result
Geometric invariants (all cases)	802/802 pass
label shape, affine, spacing match CT	—
no identifier outside the scheme	—
left/right structures correctly sided	—
Ribs present	11,548
not evaluable (expected vertebra outside the FOV)	5,788
no vertebra within the anchor distance	11
evaluable	**5,749**
matching the expected vertebra	5,747
residual offset	2 (0.035% of evaluable)
misnumbered cases	0
Spinopelvic medians within published range	6/6

TABLE II Derived measures against a published asymptotic reference ($n = 260$, standing)[1]. Sacral slope and pelvic tilt are postural where pelvic incidence is not, and this cohort is supine. The two postural measures depart from the standing reference in opposite directions — sacral slope 4.1° lower, pelvic tilt 4.3° higher — which is not two discrepancies but one, because $PI = PT + SS$ holds by construction: with pelvic incidence fixed at its published value, any fall in sacral slope must appear as an equal rise in pelvic tilt. That the two offsets match in magnitude is evidence the geometry is internally consistent rather than evidence of error.

Measure	This dataset (supine)	Reference (standing)
Pelvic incidence	54.7°	54.7 ± 10.6
Pelvic tilt	17.3°	13 ± 6
Sacral slope	36.9°	41.0 ± 8.4
Lumbar lordosis (supine)	52.7°	43 ± 11.2 (standing)
PI − LL mismatch	2.6°	centred near 0

and because more than half of the ribs in a screening abdominal CT being unevaluable is itself a fact about the corpus worth reporting. Both offsets are field-of-view truncations on individually characterised cases and are reported rather than suppressed: a threshold tuned until the count reaches zero yields a cleaner number and a less informative one.

Agreement with published values. Derived spinopelvic measures are compared against Vialle *et al.*[1] (Table II). Pelvic incidence is the strongest of these checks because it is a morphological property of the pelvis rather than a posture, and so is directly comparable to a standing reference cohort.

III. DATA FORMAT AND USAGE NOTES

Each record is a gzipped NIfTI image/label pair sharing an identical 4×4 affine, canonicalised to PIR, requiring no resampling. Masks are NIfTI rather than DICOM: the policy prefers DICOM where applicable, and for volumetric segmentation masks NIfTI is what the downstream tooling expects.

Splits are patient-grouped and LSTV-stratified five-fold with a held-out test set, frozen and shipped with the data. Stratification enforces a minimum number of lumbarisation

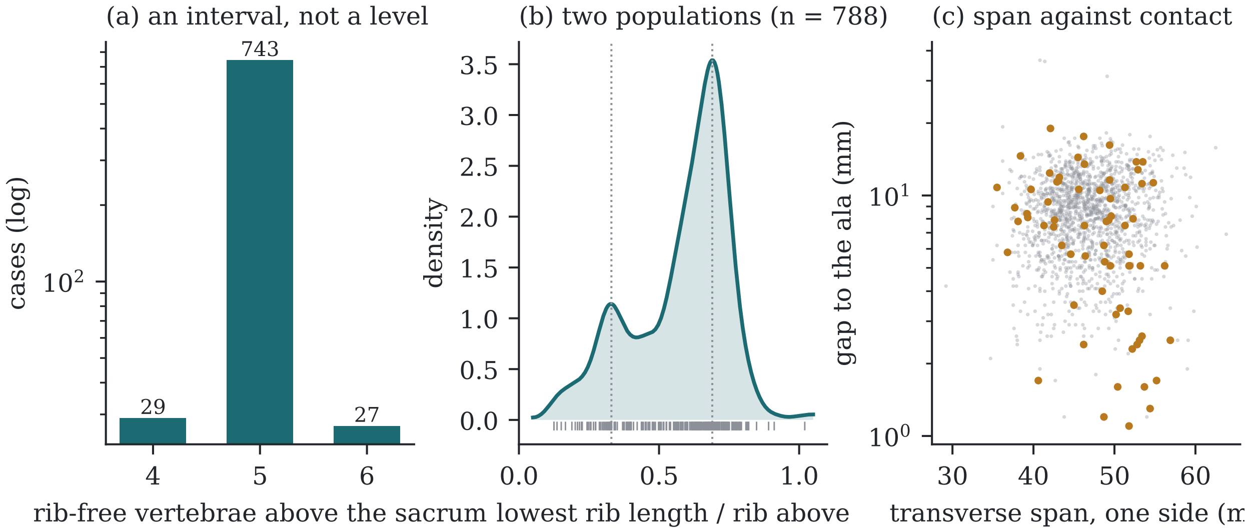


FIG. 3 Count-free measures, valid without knowing which vertebra is which. (a) Vertebrae between the lowest rib and the sacrum: an interval, not a level assignment. (b) Lowest rib length as a fraction of the rib above, showing two populations near 0.33 and 0.69 rather than one distribution with a tail. (c) Lowest lumbar transverse span against its gap to the sacral ala; cases carrying a source transitional label are marked.

cases in every validation fold, so no fold is evaluated on a cohort missing the rare class the dataset exists for.

The archive of record is Zenodo (10.5281/zenodo.XXXXXXX). Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records. 738 are recorded female (393) or male (345); 11 carry DICOM's *other* value for patient sex, which is a recorded value and not a missing one, and 53 carry no sex field at all. Age is present for 709 (median 59, range 50–89). All 802 are counted in totals, and records outside the female/male dichotomy are excluded from measures stratified by sex — a limitation of those measures rather than a property of the cohort, and stated here because folding eleven people into "unrecorded" would misdescribe what the source says. The lower age bound is the screening guideline rather than a selection applied here, and it is why the measures below sit closer

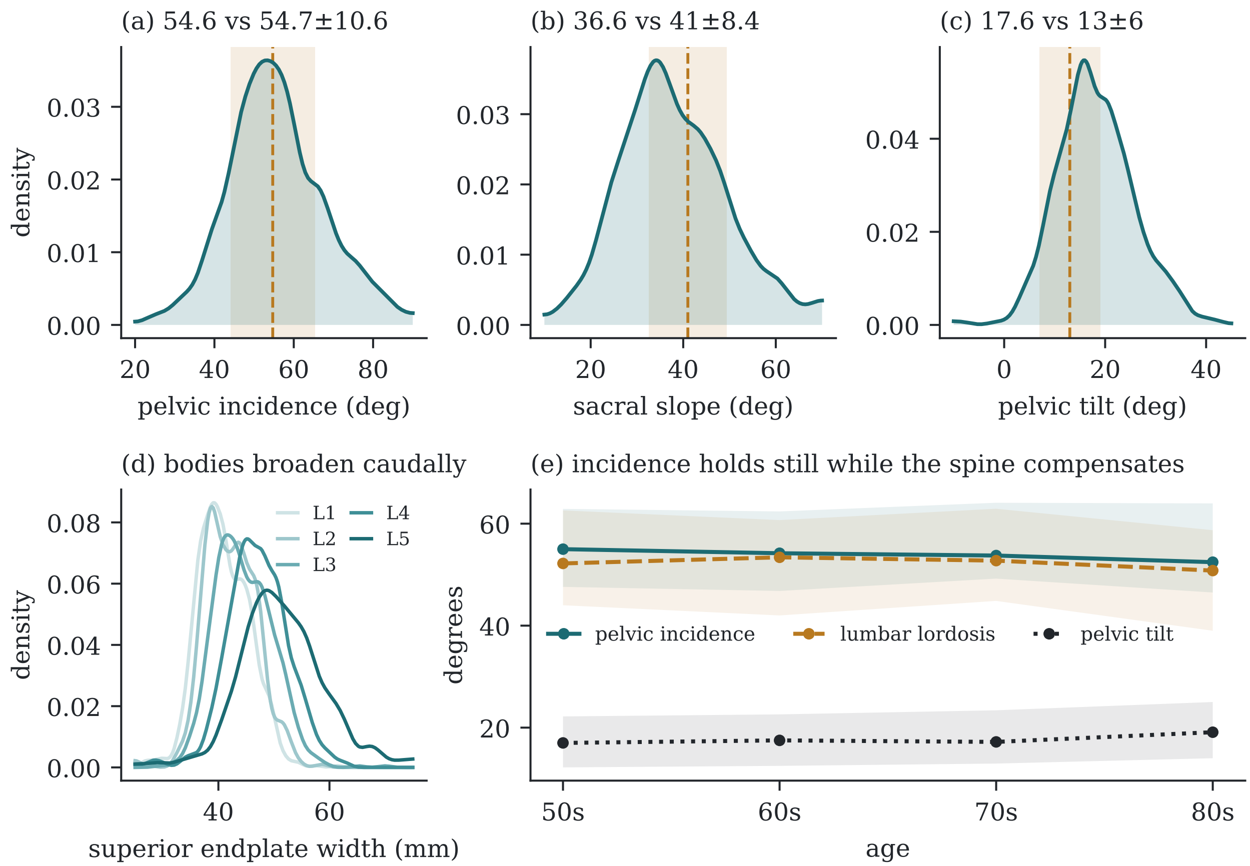


FIG. 4 (a–c) Spinopelvic measures against published reference bands (mean ± SD). (d) Superior endplate width by level. (e) Median and interquartile range by decade; pelvic incidence is flat by construction of the anatomy, which makes it a negative control for the measures beside it.

to a reference range than a surgical series would.

Count-free measures (Fig. 3). Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which of the two is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal, with modes near 0.33 and 0.69.

Level gradients and spinopelvic measures (Fig. 4). Vertebral dimensions increase caudally in step with the load each level carries. Pelvic incidence does not vary with age, while pelvic tilt increases — the descriptive signature of a pelvis retroverting as a spine ages.

Opportunistic measures (Fig. 5). Every scan carries a vertebral bone density measurement at no additional dose. L1 trabecular attenuation is reported at the published standard site.

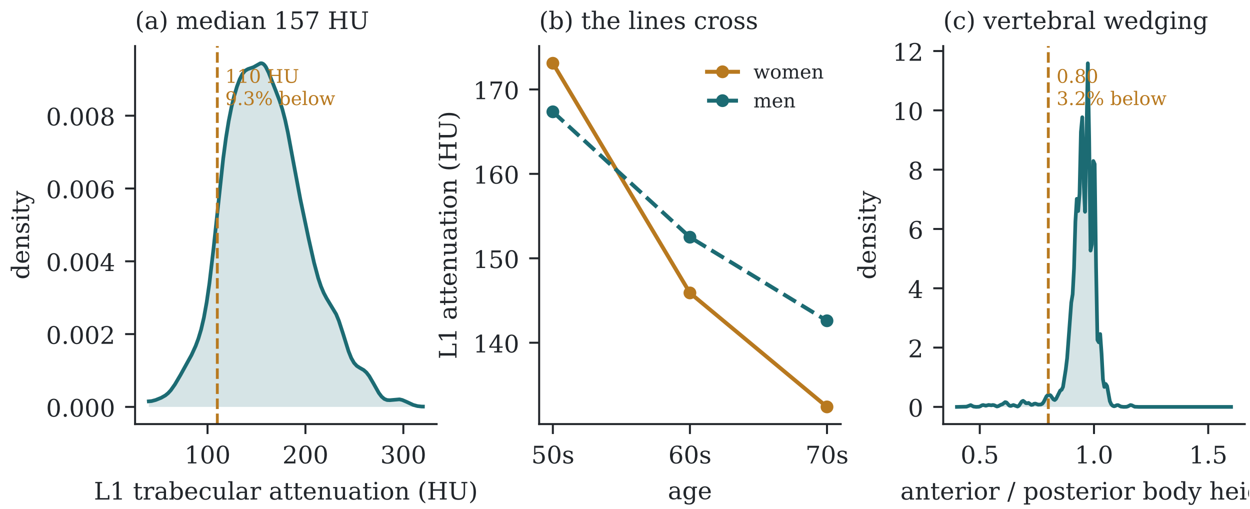


FIG. 5 Opportunistic measures from the same images. (a) L1 trabecular attenuation with the 110 HU osteoporosis-specific threshold. (b) Median attenuation by decade and sex. (c) Lowest lumbar wedge ratio with Genant[8]'s 0.80 threshold for mild wedge deformity.

V. POTENTIAL APPLICATIONS AND LIMITATIONS

The dataset supports segmentation and level-numbering benchmarks, anatomical variant studies, surgical planning research, and opportunistic screening work of the kind Fig. 5 illustrates.

Its limitations are part of it. Thoracic ground truth is field-of-view limited and does not extend to T1. Postural angles are supine; pelvic incidence is not postural and needs no such caveat. Soft-tissue (58–73) and hardware (76–79) identifiers are declared in the scheme and populated by no case in this release. Transitional labels derive from two independent sources and are not uniformly adjudicated, which is precisely why the measures reported here are count-free. Finally, the cohort is a screening population, so its distributions should not be read as representative of a surgical one.

DATA AVAILABILITY

The dataset is permanently archived at https://doi.org/10.5281/zenodo.XXXXXXX. Code is at [repository URL], tagged at the release commit.

CONFLICT OF INTEREST

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